

THE UNIVERSITY OF CHICAGO

MASTERS IN COMPUTATIONAL SOCIAL SCIENCE
THE UNIVERSITY OF CHICAGO

# MACS 30111

# M8: LAST DAY!!!!

# Announcements

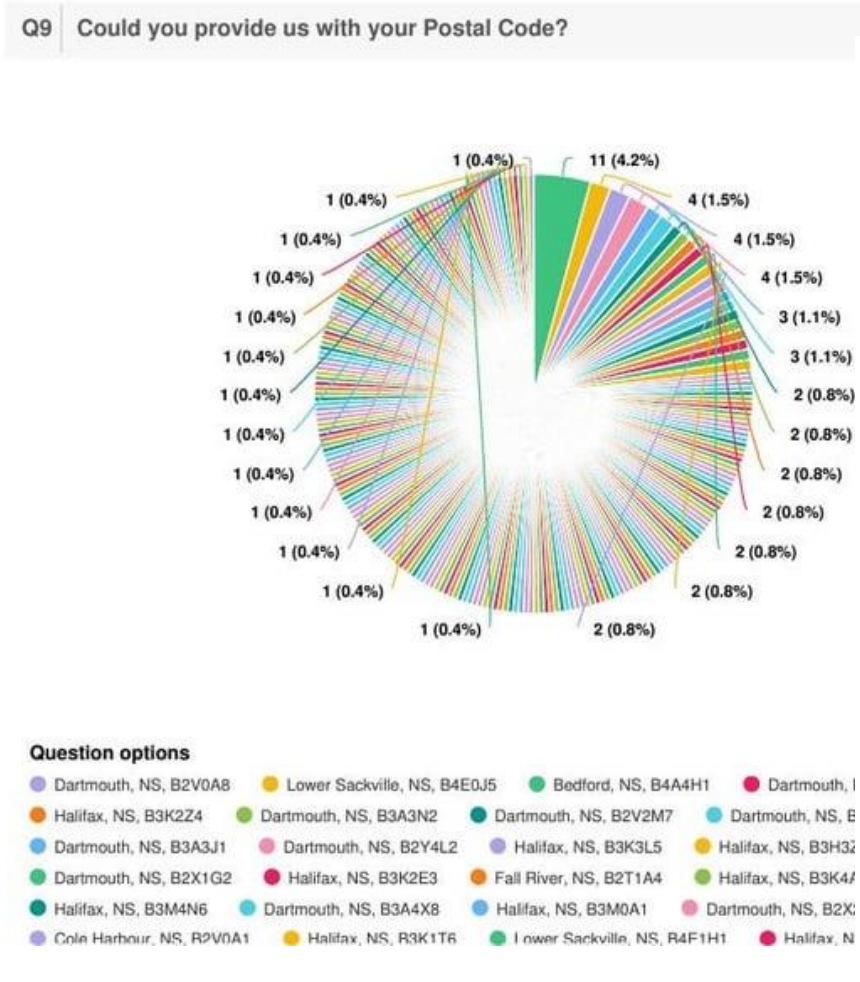
- **Switch to accelerated track: need MPCS exam**
  - https://masters.cs.uchicago.edu/student-resources/placement-exams/
  - Register, take the exam

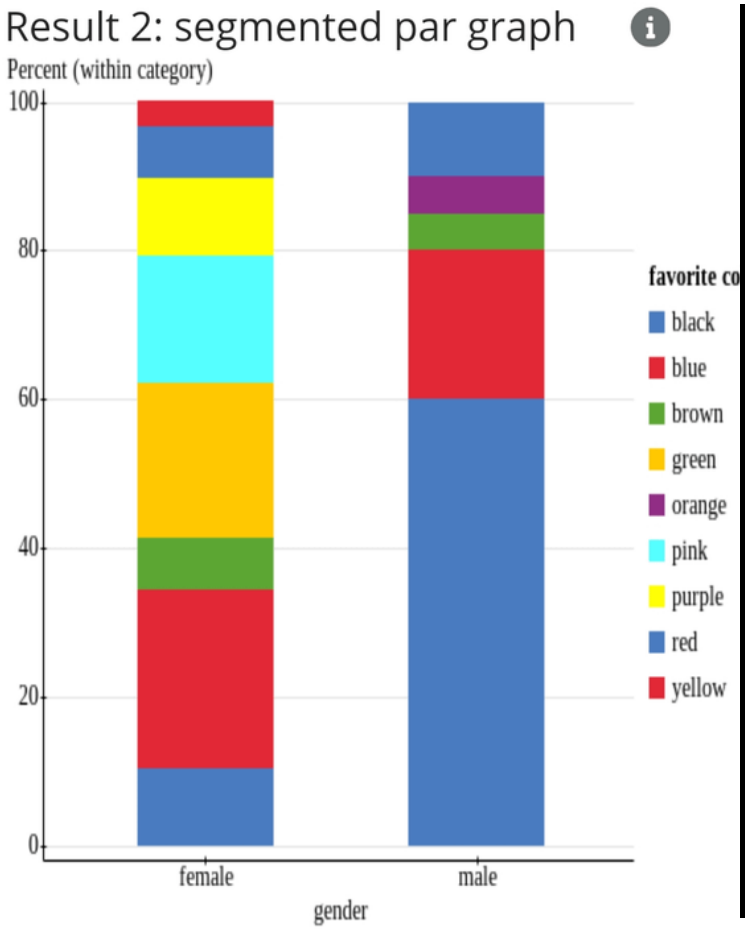
- **Final exam schedule: TWO START TIMES**
  - Tues, Dec 10th, 9am-11am
  - Tues, Dec 10th, 10am-12pm

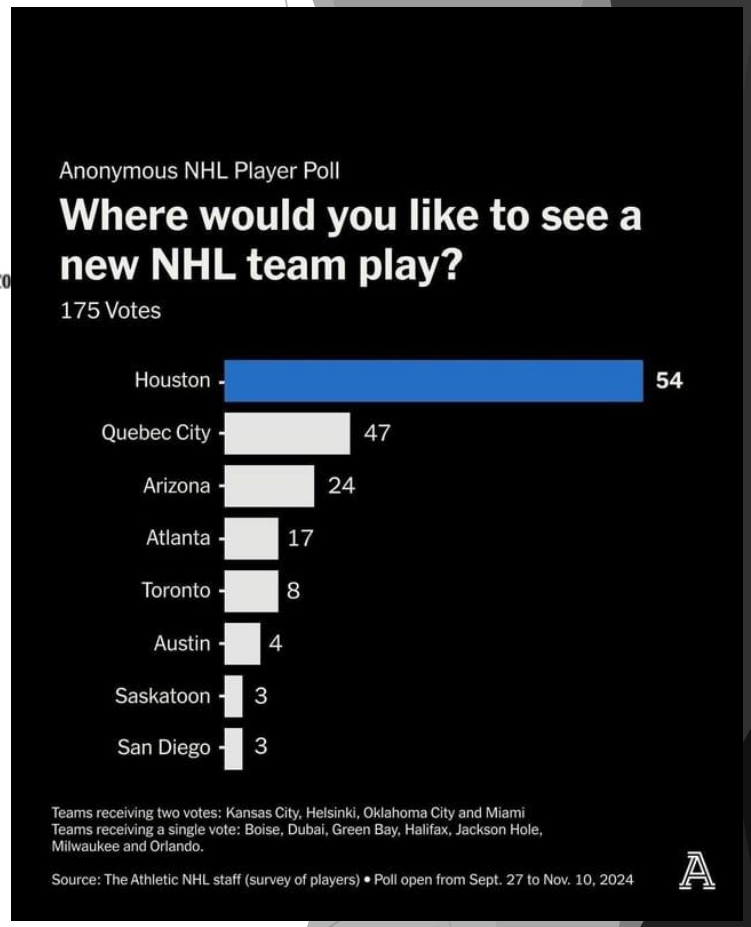
- **Review**
  - Textbook
  - Team tutorial
  - Short exercise
  - Programming assignment
  - Extra exercise

# Getting to know graphs

…but first…




https://www.reddit.com/r/dataisugly/

# Starting with a summary:

```
def tree_health_by_boro(trees):
combined_status = trees.status.where(trees.status != "Alive", trees.health)
num_per_boro = trees.groupby("borough").size()
combined_per_boro = trees.groupby(["borough",combined_status]).size()
pct_per_boro = combined_per_boro/num_per_boro*100.0
pct_per_boro_df = pct_per_boro.unstack()

return pct_per_boro_df[["Good", "Fair", "Poor", "Dead", "Stump"]]

tree_health_by_boro(trees)
```

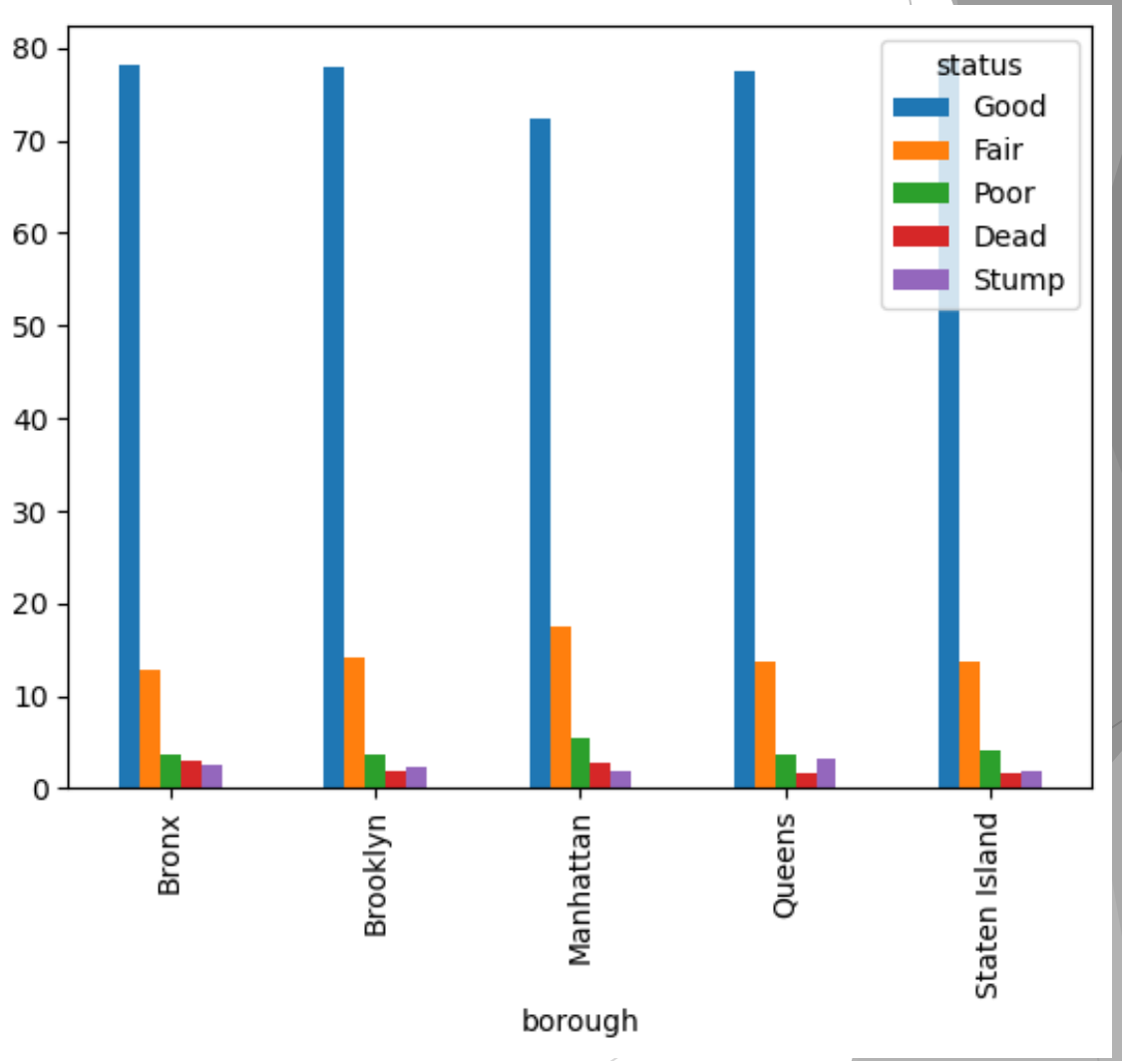
| status | Good | Fair | Poor | Dead | Stump |
|---|---|---|---|---|---|
| borough | | | | | |
| Bronx | 78.169783 | 12.777719 | 3.632501 | 2.969379 | 2.450618 |
| Brooklyn | 77.956829 | 14.142126 | 3.643122 | 1.872042 | 2.385881 |
| Manhattan | 72.387387 | 17.516775 | 5.516409 | 2.754383 | 1.825046 |
| Queens | 77.432539 | 13.789209 | 3.758516 | 1.772094 | 3.247642 |
| Staten Island | 78.494654 | 13.801060 | 4.024003 | 1.775575 | 1.903758 |

# output

# Plotting with Pandas

Using the plot method:

- tree_health_by_boro(trees).plot.bar()




Reference:

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.plot.html

# Plotting with matplotlib

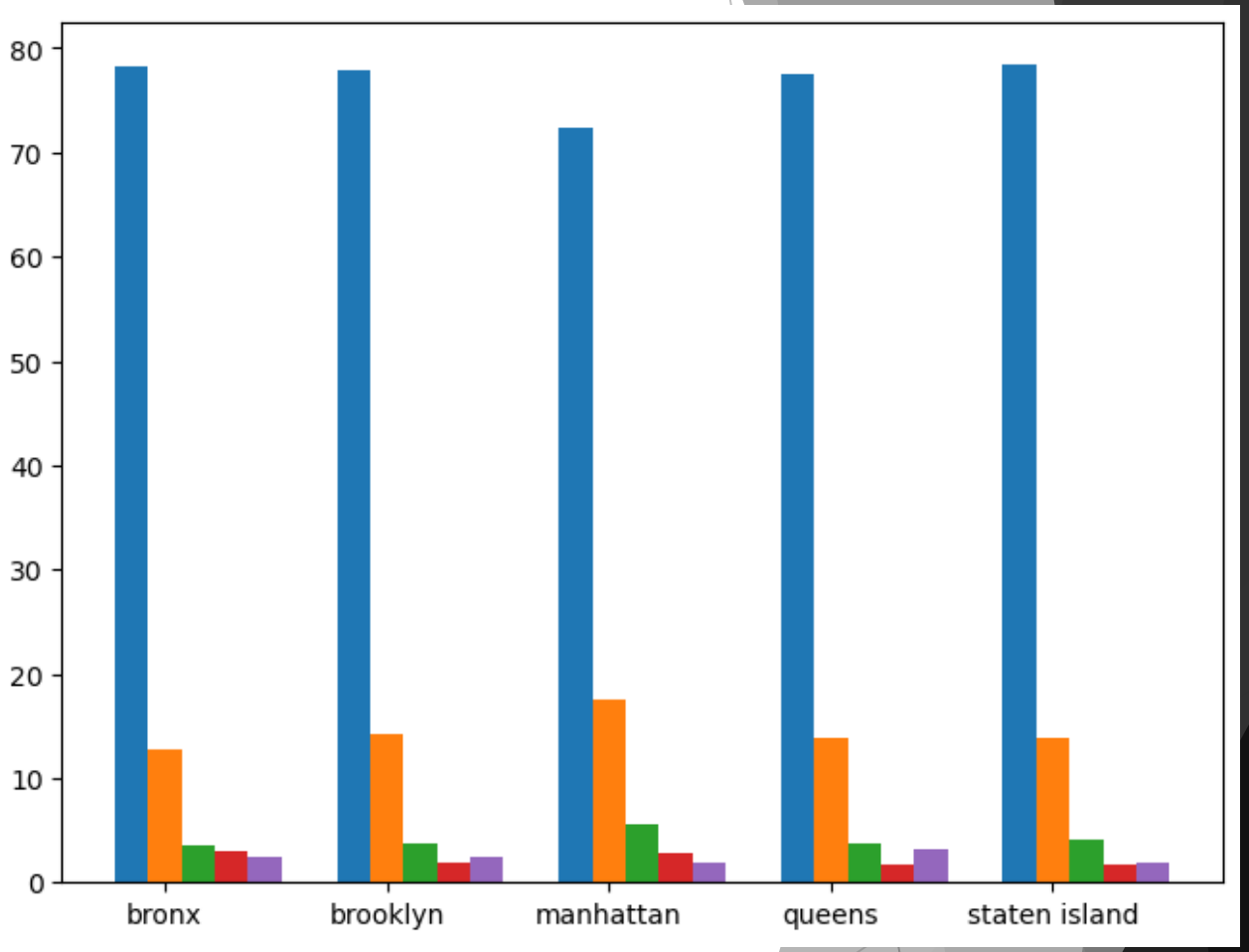
```
import numpy as np
df = pd.DataFrame(tree_health_by_boro(trees))
df.index

#df["Good"]
plt.bar( df.index, df["Good"])
plt.bar(df.index, df["Fair"])
plt.bar(df.index, df["Poor"])
plt.bar(df.index, df["Dead"])
plt.bar(df.index, df["Stump"])


x = np.arange(len(df.index)) # the label locations
width = 0.15 # the width of the bars
multiplier = 0

fig, ax = plt.subplots(layout='constrained')

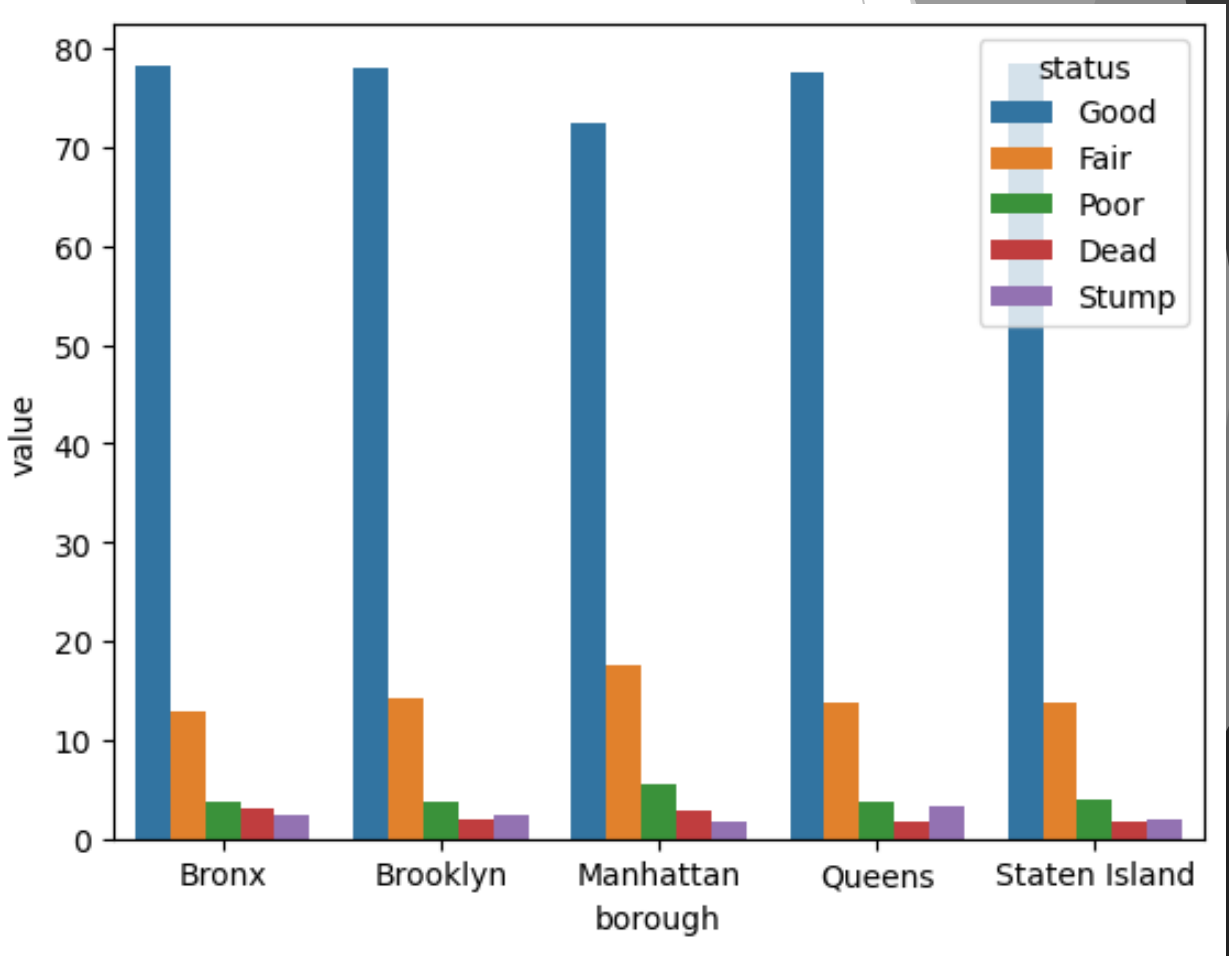
for attribute, measurement in df.items():
offset = width * multiplier
rects = ax.bar(x + offset, measurement, width, label=attribute)
multiplier += 1
ax.set_xticks(x + width, ["bronx", "brooklyn", "manhattan", "queens", "staten island"] )
plt.show()
```




https://matplotlib.org/cheatsheets/

# Plotting with Seaborn

```
import seaborn as sns
df2 = df.reset_index()
df2 = df2.melt(id_vars = "borough")
sns.barplot(df2, x = "borough", y = "value" , hue
= "status")
```




https://www.datacamp.com/cheat-sheet/python-seaborn-cheat-sheet

# Which to use?

- Probably not matplotlib…
- Usually, quick and dirty, proof-of-concept: basic plot methods but nice and pretty: **seaborn**

- PRACTICE PRACTICE PRACTICE

# Misc: other POLARS

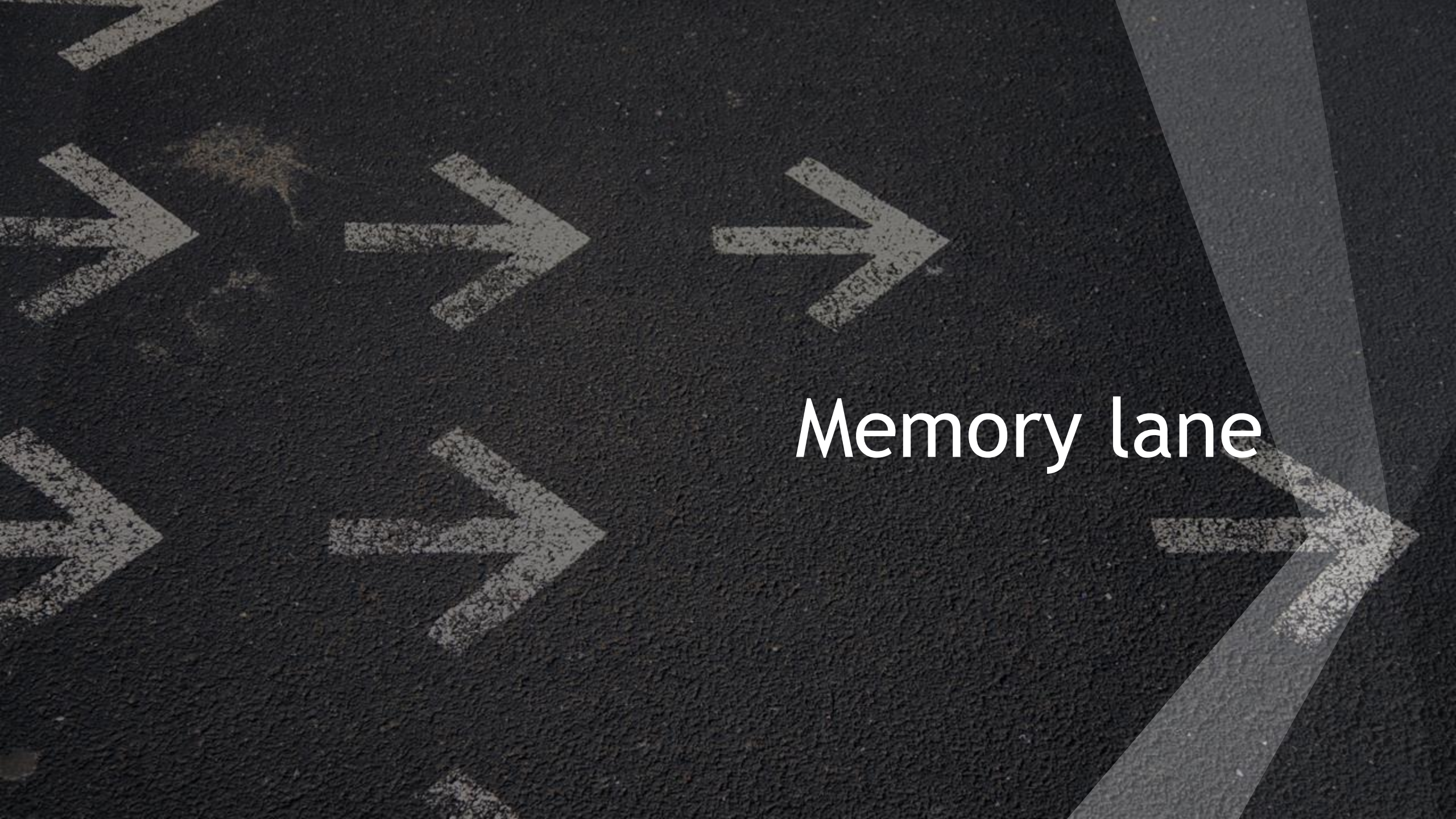



Seems cool but pandas is probably better for what you need right now

https://pola.rs/

# RECAP

- Pandas is going to be HUGE (ditto NumPy!)
- Think about what you need
- Sometimes it is MUCH faster to puzzle through and sketch before trying to do something – often there is a simpler path
- Think about what you are trying to do and what it will look like.
  - Do you need a new column?
  - Are you summarizing data?
  - Is this a ‘permanent’ or ‘temporary’ alteration?
- Graphing: google search **IMAGES**


Memory lane

# Course Description

- This course is the first in a three-quarter sequence that teaches computational thinking and skills to students in the sciences, mathematics, economics, etc. The course will cover abstraction and decomposition, simple modeling, basic algorithms, and programming in Python. Applications from a wide variety of fields serve both as examples in lectures and as the basis for programming assignments. In recent offerings, students have written programs to simulate a model of housing segregation, determine the number of machines needed at a polling place, and analyze tweets from presidential debates.

# Assignments and Skills

- Consider your bootcamp exam and your first assignment – how it felt re: time and difficulty
- Look at your most recent PA – consider how that felt re: time and difficulty!
- Tremendous progress and thoughtfulness over the term in assignments and reflections

# How we did it

- **Weekly assignments** balanced between:
  - TT – answers provided for easy comparison
  - SE – intermediate-level for slightly more challenge
  - PAs – more/most challenging but smaller number required
- **Exams:**
  - In-class to assess recall / facility with concepts (mirrors interview questions, etc.)
  - Take-home to assess ability to work on more complex projects
- **Class sessions:**
  - Applied hands-on lessons where we interact and grapple with code and problems
  - Use of Ed discussion to provide channels for communication
  - Regular and expanded OH around tricky assignments and important deadlines

# Evaluations: purpose and role

- Help us understand how your learning connected with / came from the course
- Additional (instructor-designed) questions:
  - One midterm vs two
  - Idea: two midterms + optional final. Best of three.

# Where we go from here

- EXAM (!!!)
- Final course reflection + evaluation
  - Reflection = your own growth and progress
  - Evaluation = course design and structure
- KEEP IN TOUCH!
  - I want to hear how your future projects go, what you do, the courses you take, etc!
  - Feedback: are there skills you later use that are/not helpful? In what way(s)? How can this course be improved to help you in your later coursework?